

Candidate's Name \_\_\_\_\_

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**MERIDIAN JUNIOR COLLEGE**

JC2 Preliminary Examinations

Higher 2

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**H2 BIOLOGY**

Paper 1    Multiple Choice

**9648/01****26 September 2014****1 hour 15 minutes**Additional Materials: OMR

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**READ THESE INSTRUCTIONS FIRST****Do not open this booklet until you are told to do so.**

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, civics group and register number in the spaces at the top of this page and all separate paper used.

There are **40** questions in this section. Answer **all** questions. For each question, there are four possible answers labelled **A, B, C** and **D**.Choose the **one** you consider correct and record your choice in **soft pencil** on the separate OMR answer sheet.

Each correct answer will score one mark. A mark will not deducted for a wrong answer.

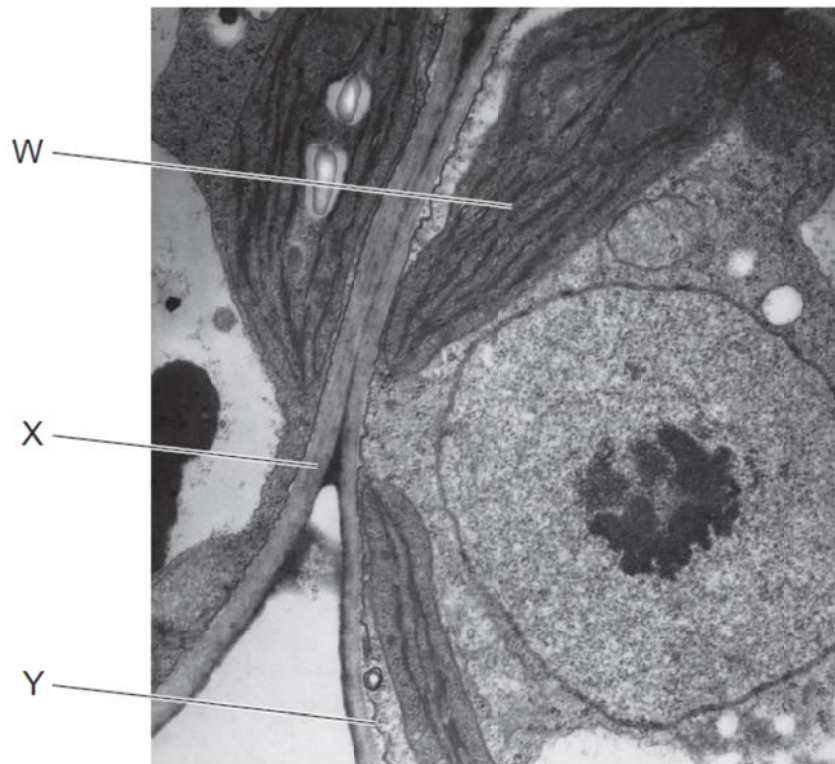
Any rough working should be done in this booklet.

Calculators may be used.

1. Which type of cell will contain the highest proportion of single membrane-bound structures?

- A Post synaptic neurone
- B  $\beta$  cell of Islets of Langerhans
- C Red blood cell
- D Smooth muscle cell

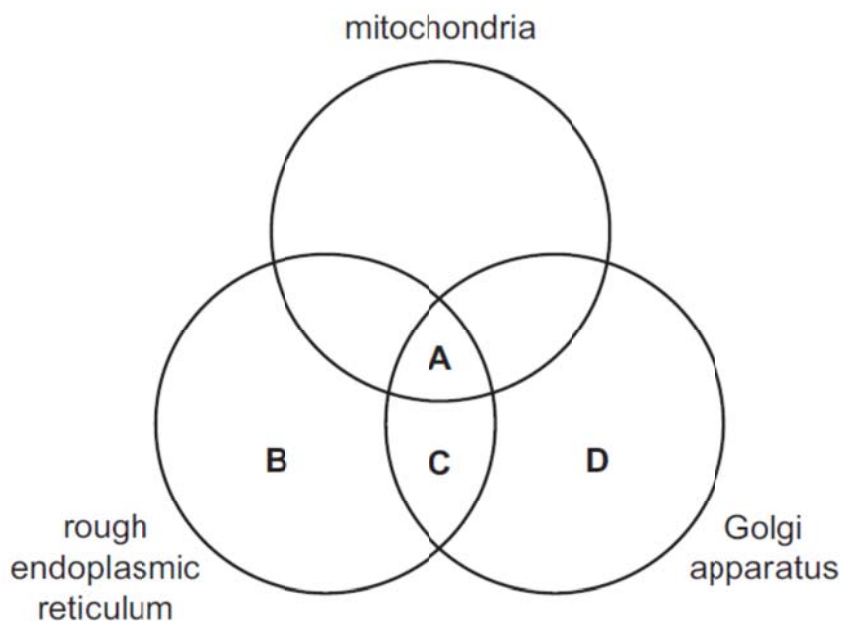
2. The electron micrograph shows part of two cells.



Which of the labelled features enable these two cells to be identified as eukaryotic?

- A X only
- B W only
- C W and X
- D W, X and Y

3. Which organelles are required in the formation of glucagon found in secretory vesicles?



4. Four different fruit juices, **A**, **B**, **C** and **D**, were tested with Benedict's solution. A second sample of each juice was hydrolysed and tested with Benedict's solution. The table shows the masses of the precipitates formed.

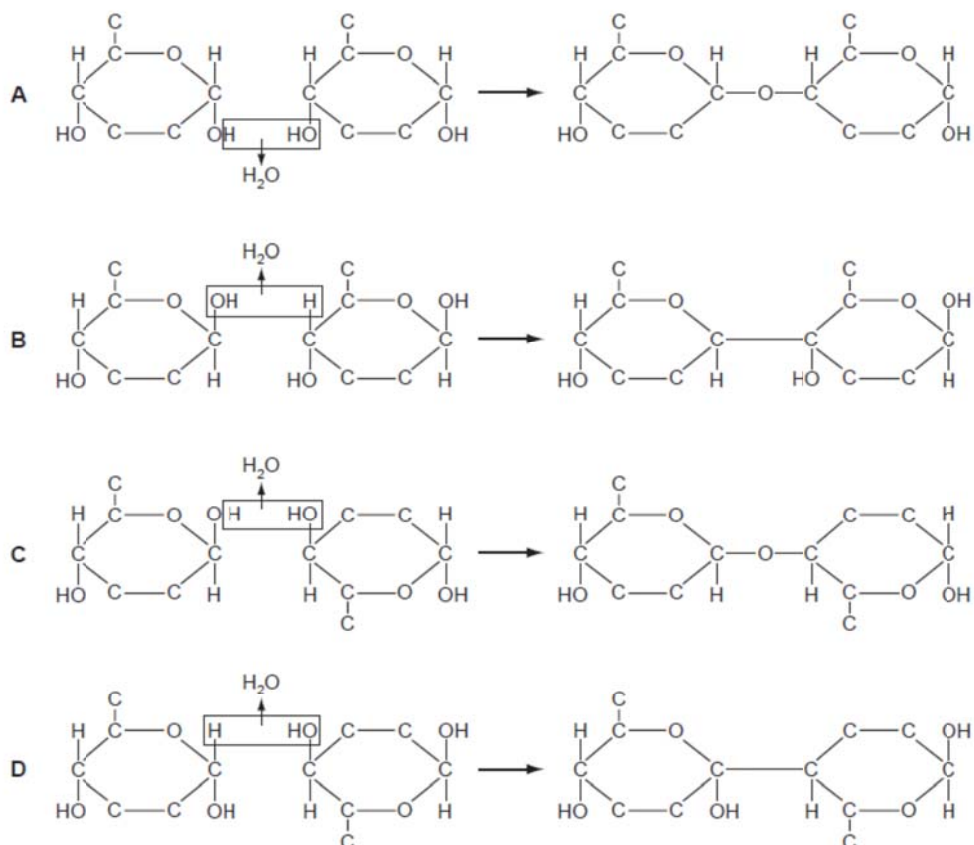
Which juice contains the greatest mass of non-reducing sugar?

|          | mass of precipitate before hydrolysis / mg | mass of precipitate after hydrolysis / mg |
|----------|--------------------------------------------|-------------------------------------------|
| <b>A</b> | 60                                         | 82                                        |
| <b>B</b> | 52                                         | 70                                        |
| <b>C</b> | 63                                         | 84                                        |
| <b>D</b> | 56                                         | 77                                        |

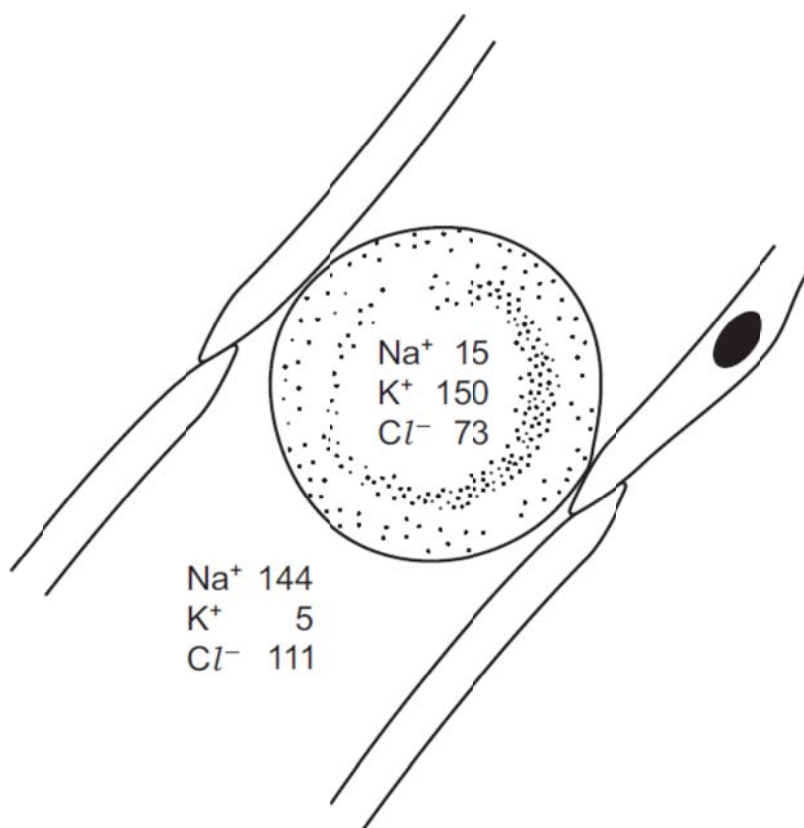
5. Which set of statements correctly describes haemoglobin?

|          |                                                                                            |                                                            |                                                                                             |                                                                          |
|----------|--------------------------------------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>A</b> | four polypeptide chains, forming a globular chain                                          | Iron ions can associate with oxygen forming oxyhaemoglobin | In each chain, hydrophilic R groups of amino acids point towards the centre of the molecule | at 50% saturation, two oxygen molecules are transported by the molecule  |
| <b>B</b> | four polypeptide chains, each containing a prosthetic group                                | iron ions can associate with oxygen forming oxyhaemoglobin | in each chain, hydrophobic R groups of amino acids point towards the centre of the molecule | at 50 % saturation, two oxygen molecules are transported by the molecule |
| <b>C</b> | four polypeptide chains, each containing a prosthetic group                                | iron ions in the molecule can bind reversibly with oxygen  | in each chain, hydrophobic R groups of amino acids surround the iron ion                    | at 50% saturation, two oxygen atoms are transported by the molecule      |
| <b>D</b> | polypeptide chains produce a loose helical shape, which curls to form a spherical molecule | Iron ions can associate with oxygen forming oxyhaemoglobin | in each chain, hydrophobic R groups of amino acids point towards the centre of the molecule | at 50% saturation, two oxygen molecules are transported by the molecule  |

6. Which diagram shows the linking of two monomers together to form cellulose?



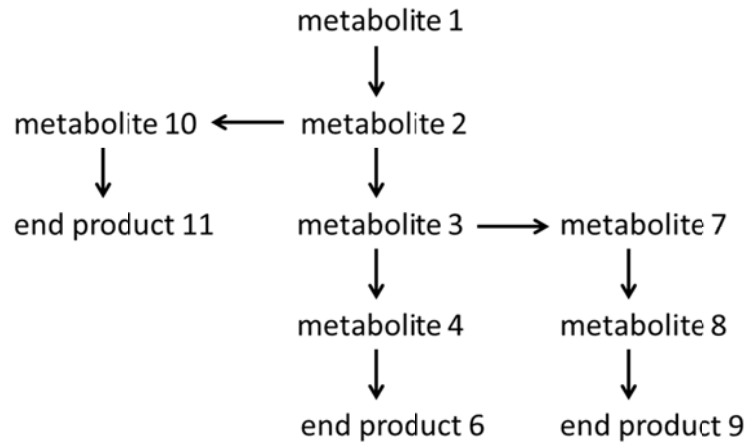
7. The diagram shows a red blood cell and the concentrations of ions, in  $\text{mmol dm}^{-3}$ , in the plasma and in the cell.



Which option correctly explains how different ion concentrations are maintained in the red blood cell?

|          | <i>into cell</i>                       | <i>out of cell</i>                    |
|----------|----------------------------------------|---------------------------------------|
| <b>A</b> | Active transport of $\text{Na}^+$      | Active transport of $\text{K}^+$      |
| <b>B</b> | Facilitated diffusion of $\text{Na}^+$ | Facilitated diffusion of $\text{K}^+$ |
| <b>C</b> | Active transport of $\text{K}^+$       | Active transport of $\text{Na}^+$     |
| <b>D</b> | Facilitated diffusion of $\text{Cl}^-$ | Active transport of $\text{K}^+$      |

8. The diagram below shows a metabolic pathway that is controlled by end product inhibition. When the concentration of the end products increase beyond a particular level, an enzyme in the pathway is inhibited to lower the rate of production.

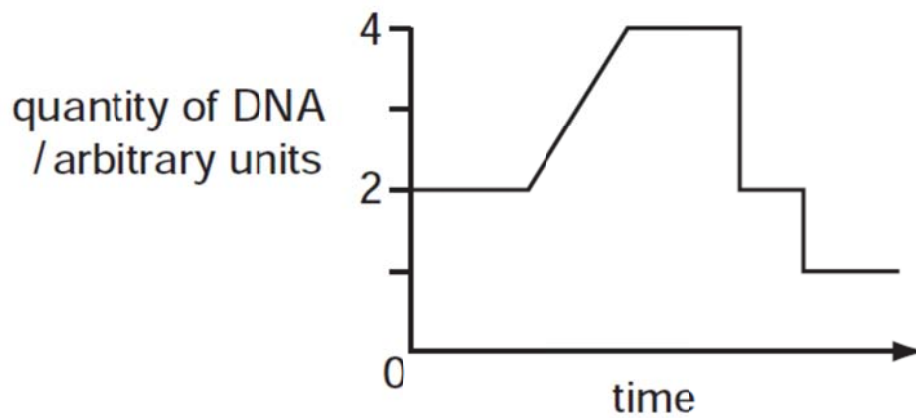


It was discovered that an increase in amount of end product 9 does not affect the synthesis of end product 11, however, it decreases synthesis of end product 6.

The enzyme controlling which of the following reactions could be inhibited by end product 9?

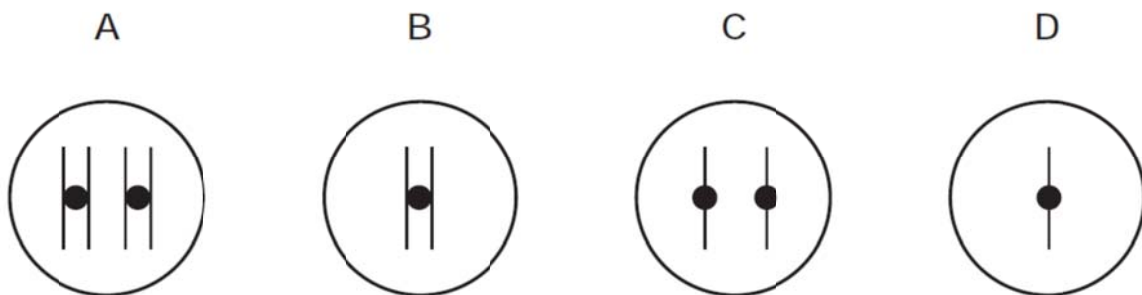
- A metabolite 1 → metabolite 2
- B metabolite 2 → metabolite 3
- C metabolite 3 → metabolite 7
- D metabolite 2 → metabolite 10

9. The graph shows the change in the quantity of DNA in a cell during a reduction division (meiosis).



A cell with one pair of chromosomes ( $2n = 2$ ) undergoes meiosis.

Which nucleus is formed as a result of this division?



10. When DNA replicates, new nucleotides containing the common isotope of nitrogen ( $^{14}\text{N}$ ) are used to build new nucleic acids.

In the laboratory, nucleotides can be synthesised using the heavy isotope of nitrogen ( $^{15}\text{N}$ ). DNA is allowed to replicate once using these  $^{15}\text{N}$  nucleotides, then twice more using  $^{14}\text{N}$  nucleotides.

What will be the ratio of  $^{14}\text{N}$  to  $^{15}\text{N}$  nucleotides in the final molecules?

- A** 1:1                      **B** 3:1                      **C** 5:1                      **D** 7:1

11. Acyclovir is a molecule that can substitute for the nitrogenous base, guanine. It is used as an antiviral drug.

Four features of acyclovir are listed below.

Which one of these features of acyclovir activity does not suggest that it could interfere with, and therefore inhibit, viral DNA replication?

- A** It can act as a competitive inhibitor of viral DNA polymerase.
- B** It can act as a non-competitive inhibitor of viral RNA polymerase.
- C** It can be incorporated into the growing viral DNA strand.
- D** It can be phosphorylated to become a nucleoside triphosphate.

12. The descriptions below are of nucleic acids in eukaryotes.

- 1 a polynucleotide of variable length formed by base pairing
- 2 a small polynucleotide with a specific three-dimensional shape
- 3 a small polynucleotide with specific sequence associated with proteins
- 4 a large polynucleotide with specific shape associated with proteins
- 5 a large polynucleotide with super-coiled sections associated with proteins

Which row matches the description to its function?

|          | carries specific amino acids | provides a site for protein synthesis | carries coded information | stores coded information | carries out transcriptional regulation |
|----------|------------------------------|---------------------------------------|---------------------------|--------------------------|----------------------------------------|
| <b>A</b> | 2                            | 3                                     | 5                         | 4                        | 1                                      |
| <b>B</b> | 2                            | 5                                     | 1                         | 3                        | 4                                      |
| <b>C</b> | 2                            | 4                                     | 1                         | 5                        | 3                                      |
| <b>D</b> | 1                            | 2                                     | 3                         | 5                        | 4                                      |

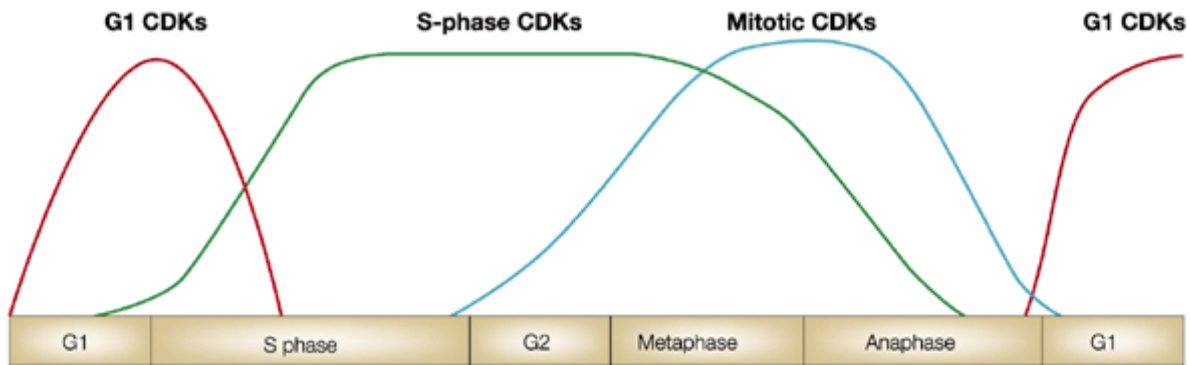
13. Below are statements that describe the control of transcription for genes encoding for enzymes in a metabolic pathway.

- 1 The genes are transcribed as a single transcription unit, with each gene having its own promoter.
- 2 The genes respond similarly to the same set of general transcription factors but respond differently to a certain set of specific transcription factors.
- 3 The genes have various combinations of control elements that enable different activators and repressors to bind and affect the rate of transcription.
- 4 The genes are found close to each other on the chromosome with chromatin remodelling as a form of transcriptional control.

Which combination of statements is true?

- A** 1 only
- B** 2 only
- C** 2 and 3
- D** 3 and 4

14. Cyclins are regulatory proteins that associate with cyclin-dependent kinases (CDKs) to control the different stages of the cell cycle. The right type and the correct amount of cyclins and CDKs must be present at the different stages to ensure regulation of the cell cycle. The changes in concentrations of the different CDKs are shown in the diagram.



Which of the following could possibly regulate the levels of the different CDKs in an actively dividing cell?

- 1 Formation of heterochromatin
- 2 Binding of repressor to promoter
- 3 Length of mRNA poly(A) tail
- 4 Tagging by ubiquitin

A 1 and 2                      B 2 and 3                      C 2 and 4                      D 3 and 4

15. Which statement about cancer is incorrect?

- A The risk of cancer is higher when *p53* gene is translocated to another chromosome having a strong promoter.
- B Mutations in proto-oncogenes may not necessarily result in excessive cell proliferation.
- C Two recessive alleles for a tumour suppressor gene must be present.
- D The growth of new blood vessels is a fundamental step in the development of a malignant tumour.

16. Which of the statements are false about RNA viruses such as influenza and HIV?
1. The virus must encode either an RNA or a DNA polymerase.
  2. The virus must contain a RNA polymerase.
  3. The virus must encode receptor protein for entry into host cells.
  4. The virus must encode proteases which can generate functional proteins from a polyprotein.
- A** 1 and 3
- B** 1 and 2
- C** 2 and 4
- D** 2, 3 and 4
17. Which of the following statements explains how the process of bacteriophage transduction in bacteria resulted in genetic variation in a population of bacteria?
- A** Bacterial proteins transferred from the donor bacterium by the phage to the recipient bacterium recombine with genes on the recipient's chromosome.
- B** The genetic material from the donor bacterium coding for phage proteins is integrated into chromosome of recipient bacterium to synthesize the corresponding proteins.
- C** The phage infection of the recipient bacterium and the introduction of DNA carried by the phage cause increased random point mutations of the bacterial chromosome.
- D** Recipient bacterial chromosome undergoes recombination with genetic material introduced by the phage from the donor bacterium.

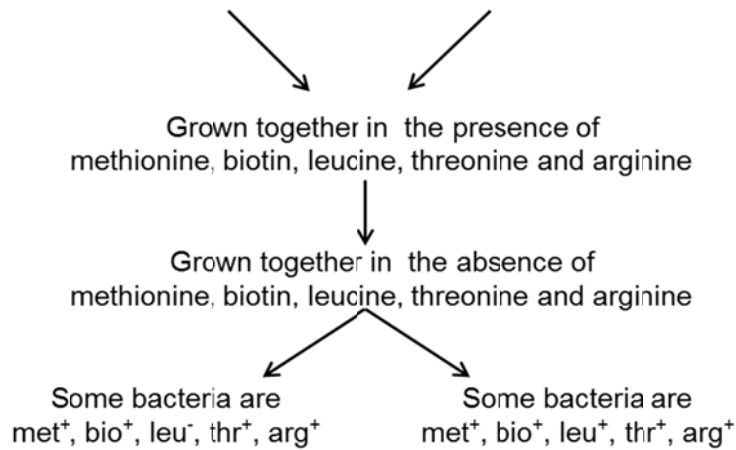
18. The diagram shows an investigation into bacterial genetics.

To grow mutant bacteria **X**:

- Need methionine (met<sup>-</sup>), biotin (bio<sup>-</sup>) and leucine (leu<sup>-</sup>)
- Do not need threonine (thr<sup>+</sup>), arginine (arg<sup>+</sup>)

To grow mutant bacteria **Y**:

- Do not need methionine (met<sup>+</sup>), biotin (bio<sup>+</sup>) and leucine (leu<sup>+</sup>)
- Need threonine (thr<sup>-</sup>), arginine (arg<sup>-</sup>)



Which process or processes could explain these results?

- 1 conjugation
- 2 transduction
- 3 transformation

**A** 1 only

**B** 3 only

**C** 1 and 2

**D** 1 and 3

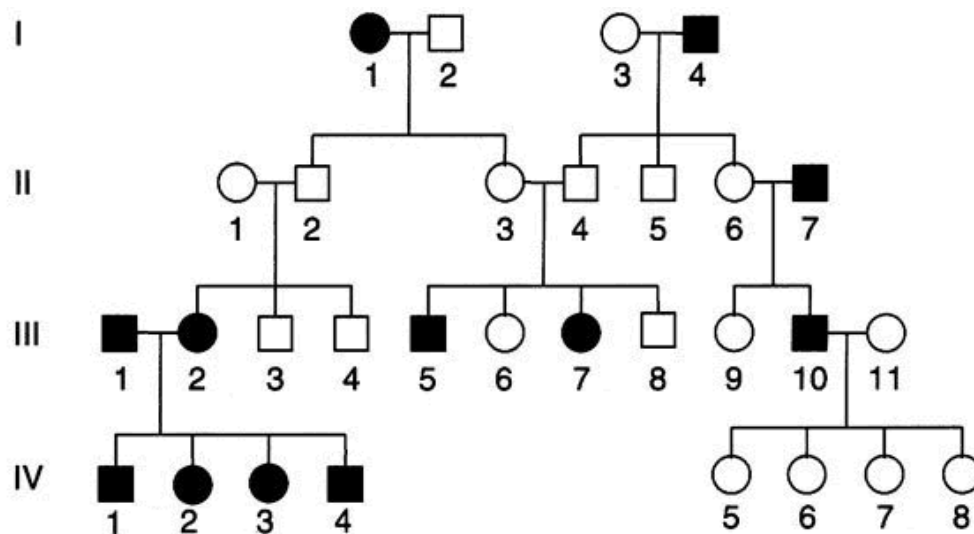
19. A tall plant is crossed with a short plant, and the progeny are all intermediate in size between the two parental plants. The intermediate F<sub>1</sub> progeny were allowed to self-pollinate, and the F<sub>2</sub> progeny were also intermediate in size, but following a normal distribution. The results of the F<sub>2</sub> progeny is best explained by

- A Environmental factors
- B Multiple alleles
- C Additive effect of polygenes
- D Epistasis

20. In a genetic cross, a plant with genotype AaBb ( $F_1$ ) was crossed with another plant with genotype aabb. In the  $F_2$  progeny, the genotypes Aabb and aaBb were more frequent than that of AaBb and aabb. The genotypes of the pure-bred parental generation should be

- A AaBb x AaBb
- B AaBb x aaBb
- C AAbb x aaBB
- D AABB x aabb

21. In the pedigree chart below, the trait, adherent earlobes, were studied.



The genotype(s) of III-3 and III-6, and the inheritance pattern are

|          | <b>Genotype(s) of III-3 and III-6</b> | <b>Inheritance Pattern</b> |
|----------|---------------------------------------|----------------------------|
| <b>A</b> | AA or Aa                              | Sex-linked recessive       |
| <b>B</b> | aa                                    | Autosomal dominant         |
| <b>C</b> | aa                                    | Sex-linked dominant        |
| <b>D</b> | AA or Aa                              | Autosomal recessive        |

22. What happens to the reduced NAD molecules in cell metabolism?

- A They act as oxidising agents in glycolysis.
- B They are oxidised in thylakoid membrane for ATP formation.
- C They are oxidised in the Calvin cycle when react with glycerate-3-phosphate.
- D They are oxidised when react with pyruvate.

23. An experiment was conducted to investigate respiration of yeast cells.

**Tube 1:** Radioactive glucose solution + suspension of yeast cells + oxygen

**Tube 2:** Radioactive glucose solution + suspension of yeast cells + oxygen + antimycin

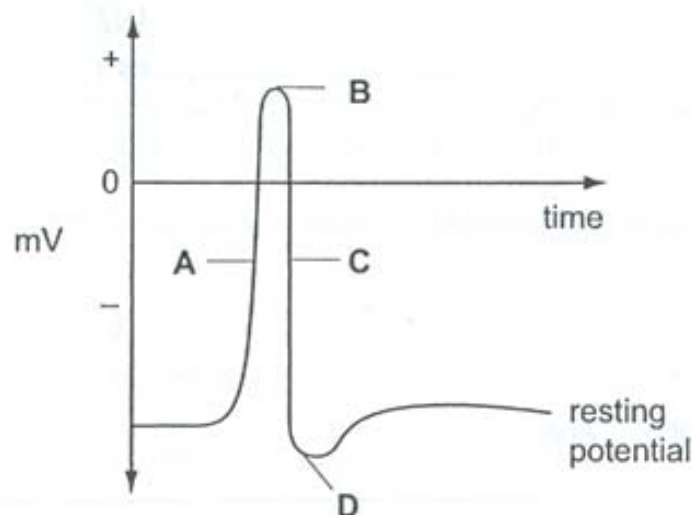
All the six carbon atoms of the radioactive glucose are  $^{14}\text{C}$ . The initial radioactivity measured in each test tube is 60 arbitrary units.

Antimycin is an electron transport chain inhibitor.

After all the glucose has been metabolized, the amount of radioactivity in the gaseous product and the content of the tubes were measured. Which of the following shows the expected result?

|          | tube 1<br>(radioactivity / arbitrary units) |                 | tube 2<br>(radioactivity / arbitrary units) |                 |
|----------|---------------------------------------------|-----------------|---------------------------------------------|-----------------|
|          | Content in tube 1                           | gaseous product | Content in tube 2                           | gaseous product |
| <b>A</b> | 40                                          | 20              | 0                                           | 60              |
| <b>B</b> | 0                                           | 60              | 20                                          | 40              |
| <b>C</b> | 0                                           | 60              | 40                                          | 20              |
| <b>D</b> | 60                                          | 0               | 40                                          | 20              |

24. The diagram shows an action potential. Which of the following statements is incorrect?



- A** At point A, the axon membrane is most permeable to sodium ions.
- B** At point B, the peak of the action potential can be reduced by decreasing the external  $\text{Na}^+$  concentration.
- C** At point C, the voltage-gated sodium channels are closed and potassium channels are open.
- D** At point D, sodium conductance changes more slowly than does potassium conductance.

25. Botulinum toxin causes degradation of proteins required for fusion of vesicles containing acetylcholine at the presynaptic membrane.

Pyrethroids in insecticides prevent the closure of the voltage-gated sodium channels in the axonal membranes.

Curare is a plant toxin that competitively and reversibly binds to the acetylcholine receptor found at the neuromuscular junction.

Which of the statements are incorrect?

- 1 Both botulinum and curare cause muscles paralysis by decreasing the generation of action potential at the cell body of neurone.
- 2 Pyrethroids cause axonal membrane to be permanently depolarized thereby resulting in convulsion in the organism.
- 3 Pyrethroids increase the frequency of action potential along axon.
- 4 Inhibition of curare can be overcome by increasing the amount of acetylcholinesterase at the synapse.

- A** 1 and 2  
**B** 2, 3 and 4  
**C** 1, 3 and 4  
**D** 3 and 4

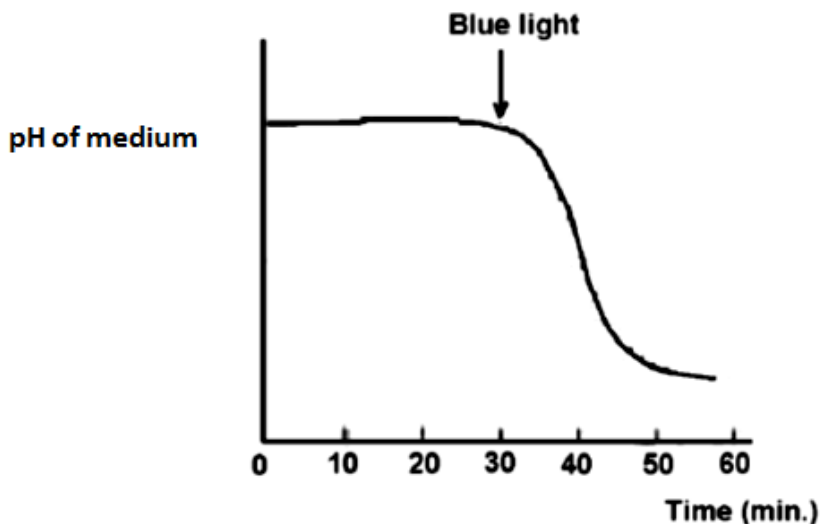
26. Saltatory conduction is faster than non-saltatory conduction because

- 1 myelinated neurons have a lower internal resistance than do non-myelinated ones, thereby leading to faster conduction.
- 2 the insulation provided by myelin sheath forces the depolarizing current farther down the axon, thereby allowing the action potential to skip parts of the membrane.
- 3 myelinated axons are bigger than non-myelinated ones, and larger diameter axons conduct faster.
- 4 there are higher concentration of sodium channels at the nodes which generates much larger action potential than the non-myelinated axon, thus, generate faster conduction.

Which of the statement(s) is/are correct?

- A** 1 and 2  
**B** 3 and 4  
**C** 2 only  
**D** 4 only

27. Which of the following statement best illustrates the outcome of the negative feedback loop in a homeostatic mechanism?
- A The rate of glycogenesis in liver cells increases in response to insulin.
  - B Glucagon is released by the pancreas in response to a drop in blood glucose concentration below the normal level.
  - C Blood glucose level increases to the normal level in response to the release of insulin.
  - D Release of glucagon is stopped in response to blood glucose concentration returning to the normal level.
28. The result of an experiment which uses guard cell protoplasts of *Vicia faba* is given below. Protoplasts (plant cells without cell wall) were incubated in a suspension medium with isotonic osmotic pressure. After 30 min under saturating red light they were irradiated with blue light for 30 sec. During the experiment in which the protoplasts were cultured, the pH of the medium was monitored.



What would be the most plausible conclusion based on the above results?

- A Red light may help guard cells to take up protons from outside into the cell.
- B Blue light may enhance the ability of guard cells to pump protons out of the cell.
- C Blue light may be a very effective wavelength of light for the respiration of the guard cells.
- D Other wavelengths of light may help guard cells to transfer protons.

29. Plant chloroplasts are believed to have evolved from cyanobacteria-like organisms through endosymbiosis. Genes in the cyanobacterial ancestor are retained in chloroplast genomes. Which of the following statements support this hypothesis?

- I. Chloroplasts and cyanobacteria share similar photosynthetic pigments and thylakoid membranes.
- II. Cyanobacteria exhibit photosynthesis.
- III. Chloroplasts and cyanobacteria share similar DNA and 80S ribosomes.
- IV. Prokaryotic genes can be expressed in chloroplasts.

- A** I and II only
- B** I, II and IV
- C** I and III only
- D** I, II, III and IV

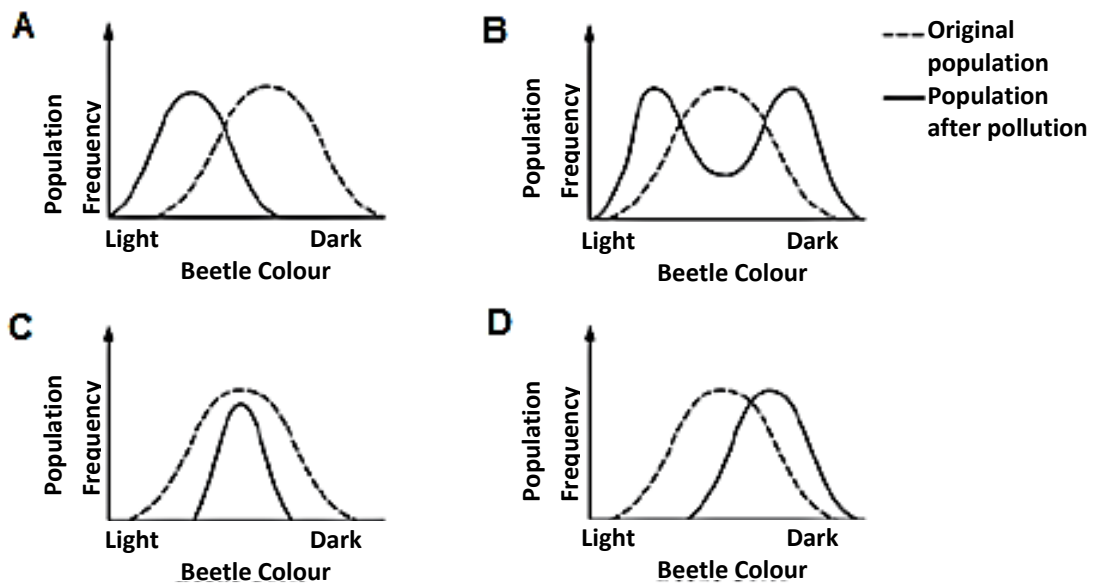
30. Which of the following statements is/are true for both genetic drift and natural selection?

- I. Both mechanisms lead to speciation.
- II. Both require variation.
- III. Both result in adaptation.
- IV. Both affect the gene pool of the population.

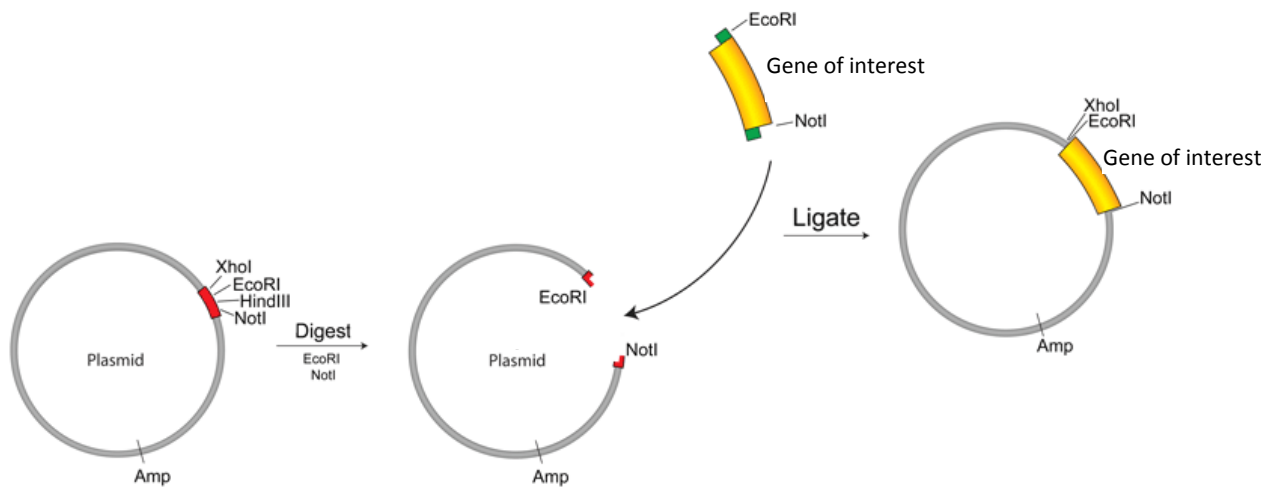
- A** I and II
- B** I and IV only
- C** II and III only
- D** IV only

31. In a hypothetical population of beetles, there is a wide variety of colour, matching the range of coloration of the tree trunks on which the beetles hide from predators. The graphs below illustrate four possible changes to the beetle population as a result of a change in the environment due to pollution that darkened the tree trunks.

Which of the following graphs includes the most likely change in the coloration of the beetle population after pollution?



32. Diagram below shows a process of genetic engineering.



All of the following statements incorrectly explain the purpose of the process, except

- A The gene of interest was inserted into the genetic marker to allow for identification of bacteria with recombinant plasmid.
- B The recipient plasmid was isolated from a bacteria and grown in mass to ensure success of this process
- C The use of two separate restriction enzymes to cut the plasmid and gene of interest allows the gene of interest to be inserted in a specific orientation.
- D EcoRI and NotI restriction sites are chosen to prevent insertional inactivation of Amp gene.

33. Which statement is incorrect?

- A cDNA method is used to map out the complete human genome
- B There is no promoter sequence in clones created from cDNA libraries.
- C To create a cDNA library, DNA ligase and a RNA-dependent DNA polymerase is required.
- D Non-coding sequences are not expressed in cDNA libraries.

34. Through the use of tissue culture techniques, genetic engineering can be more readily employed to create genetically modified plants. This is because

- A The issues surrounding GMO are animal-related.
- B A single genetically modified plant cell can developed into a entire plant
- C There are more known techniques to genetically modify plant cells
- D Plants are usually more resilient and can survive the harsh manipulation of genetic engineering

35. Which of the 2 primers can be used to amplify the sequence shown below by Polymerase Chain Reaction?

5' – GGAATTCGT -- // -- TATGCTACC – 3'

- A 5' GGTAGCATA 3' & 5' GGAATTCGT 3'
  - B 5' ACGAATTCC 3' & 5' TATGCTACC 3'
  - C 5' TATGCTACC 3' & 5' GGAATTCGT 3'
  - D 5' ACGAATTCC 3' & 5' GGTAGCATA 3'
36. Identification of individuals predisposed to a disease is done by performing DNA analysis techniques such as Polymerase Chain Reaction and gene probing after Southern Blotting. PCR requires the presence of primers for amplification of target gene while a gene probe is used to identify gene of interest.

Which of following comparison of gene probe and primers are false?

- I. Both are short and single stranded nucleic acids.
- II. Both have sequences that are complementary to 3' end of the target gene.
- III. Both can only be made up of deoxyribonucleotides.
- IV. Both must be radioactively-labelled to be identified by autoradiography.

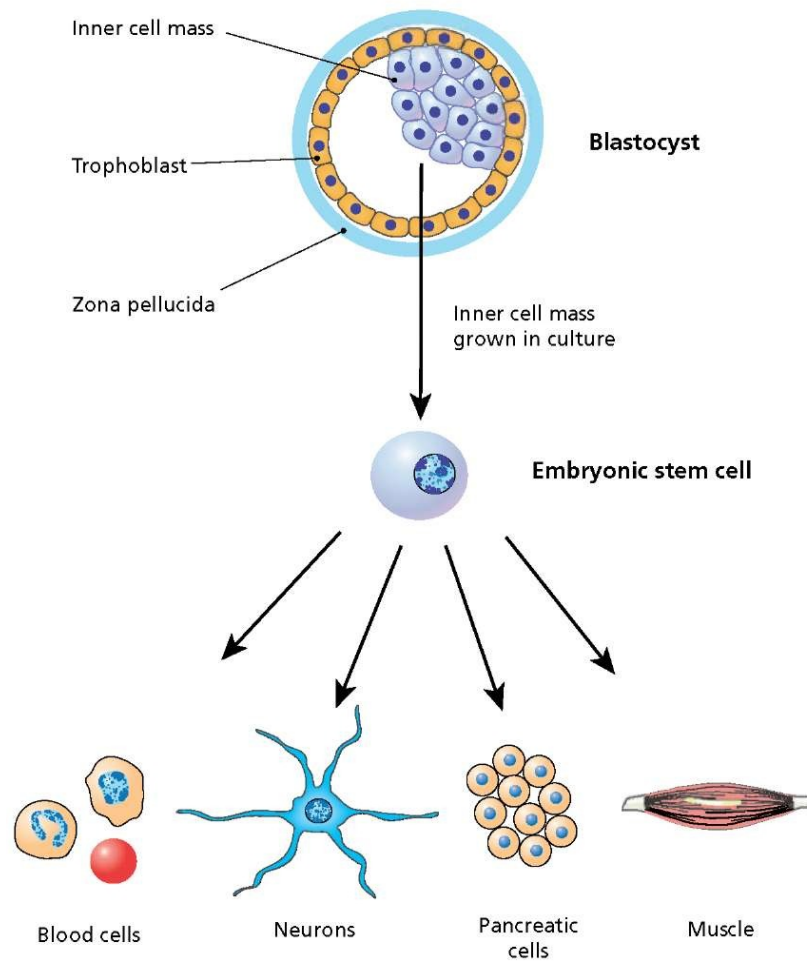
- A I and II
  - B II and IV
  - C III and IV
  - D II, III and IV
37. Some stages of a procedure involved in paternity testing are listed.

- 1 Alkali denatures DNA
- 2 Fragments are migrates at different rate depending on the number of base pairs
- 3 Probes bind to DNA which can be identified.
- 4 DNA samples are obtained from different individuals
- 5 Fragments are transferred from agarose gel to nylon membrane.
- 6 Several STR loci are isolated using suitable restriction enzymes

Which sequence shows the correct order of these stages?

- A 6 → 4 → 2 → 5 → 1 → 3
- B 6 → 4 → 2 → 1 → 5 → 3
- C 4 → 6 → 2 → 5 → 1 → 3
- D 4 → 6 → 2 → 1 → 3 → 5

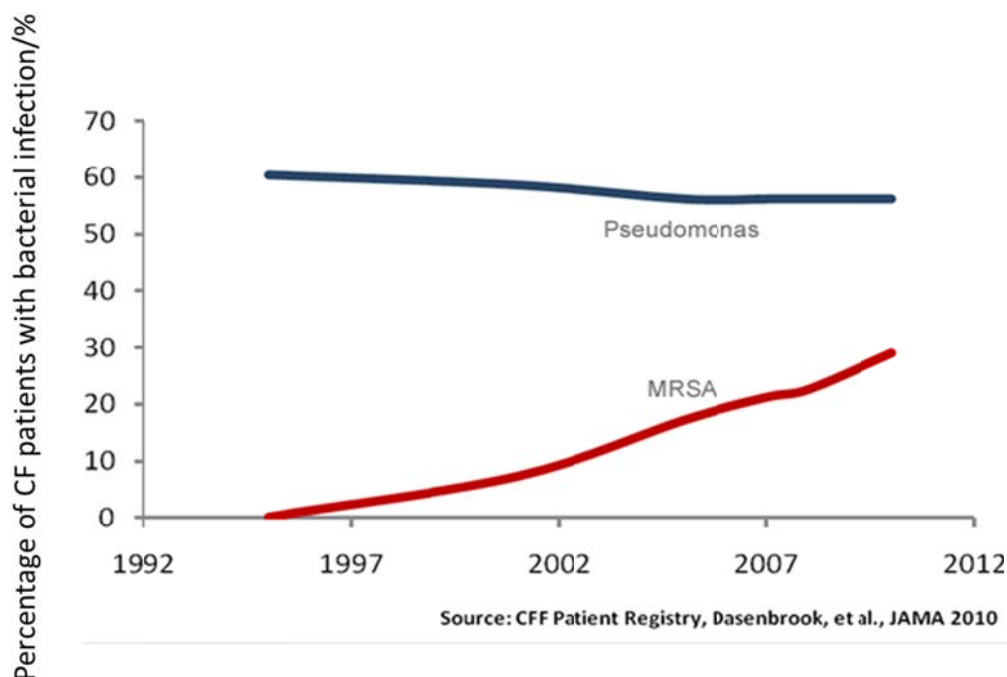
38. Which of the following features of the embryonic stem cells and specialized cells shown in the diagram are true?



|          | Embryonic stem cells                                                                                         | Specialized cells                                                                                                       |
|----------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>A</b> | Embryonic stem cells display greater plasticity when grown in culture than when in blastocyst.               | The blood cells are genetically different from the embryonic stem cells but have shorter telomeres.                     |
| <b>B</b> | Embryonic stem cells are pluripotent and are capable of transdifferentiating into many different cell types. | The pancreatic cells are genetically identical to the embryonic stem cells but with a different set of genes expressed. |
| <b>C</b> | Embryonic stem cells are multipotent and are capable of differentiating into limited range of cell types.    | The blood cells are genetically different from the embryonic stem cells because different genes are expressed.          |
| <b>D</b> | Embryonic stem cells are pluripotent and are capable of differentiating into many different cells types.     | The pancreatic cells are genetically identical to the embryonic stem cells but have shorter telomeres.                  |

39. Cystic fibrosis (CF) is a genetic disease characterized by thick, sticky mucus in the lungs and chronic lung infections resulting in gradual loss of lung function. The most prevalent lung pathogen in CF patients is *Pseudomonas aeruginosa*, which is commonly treated using inhaled antibiotics.

In recent years, methicillin-resistant *Staphylococcus aureus* (MRSA), a bacterium that is resistant to conventional antibiotics has become increasingly common. Persistent MRSA infections have been found to cause a decline in lung function and are associated with a significantly shortened life expectancy. Currently there is no approved inhaled treatment for MRSA infections in CF patients.



Which of the following conclusions can be derived from the information provided?

- I. The inhaled antibiotics against *Pseudomonas* are ineffective which results in the high percentage of CF patients infected with *Pseudomonas*.
- II. The inhaled antibiotics against MRSA are ineffective which result in an increase in percentage of CF patients infected with MRSA.
- III. The thick mucus in CF obstructs the pancreatic duct and prevents pancreatic enzymes from reaching the small intestines to digest the bacteria.
- IV. *Staphylococcus aureus* is likely to contain gene that codes for intracellular proteins that reduce antibiotic inhibitory effect.

- A II only
- B IV only
- C I, II and IV only
- D II, III and IV only

40. Some of the limitations, ethical and social concern/s of gene therapy are listed below.

1. The definition of normal and disability lies in the decision of a group of scientists.
2. Multiple rounds of gene therapy may be required as vectors do not integrate normal allele into host genome.
3. Viral vectors have known to produce harmful side effects.
4. The presence of a cure might demean the lives of individuals presently affected by disabilities.
5. The formation of class distinctions when gene therapy is only available to individuals with financial means.
6. Modification of germlines affects subsequent generations.

Which row identified the limitations and the types of concerns?

|          | <b>Limitations</b> | <b>Ethical concern/s</b> | <b>Social concern/s</b> |
|----------|--------------------|--------------------------|-------------------------|
| <b>A</b> | 1 and 3            | 2, 5 and 6               | 4                       |
| <b>B</b> | 2 and 3            | 1, 4 and 6               | 5                       |
| <b>C</b> | 2 and 4            | 1 and 6                  | 3 and 5                 |
| <b>D</b> | 1, 2 and 3         | 4 only                   | 5 and 6                 |

**END-OF-PAPER**